Comparing EEG-preprocessing pipelines in infants: A test-retest analysis of neural tracking

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INTRODUCTION

Neural tracking = alignment of brain oscillations with speech.

Speech–brain coherence (SBC) quantifies neural tracking of speech, and predicts later language development (Çetinçelik 2023; Menn, Ward et al. 2022).

The YOUth cohort provides longitudinal EEG data from ~2,000 Dutch children (5 months, 10 months, and 2–5 years) listening to Dutch nursery rhymes.

How reliable is neural tracking (SBC) in 10-m-olds?
Can we automate an EEG pre-processing pipeline?

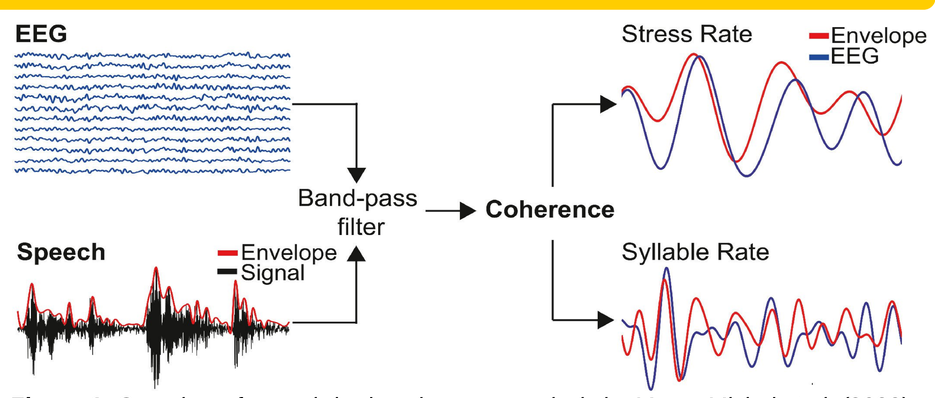


Figure 1: *Overview of speech-brain coherence analysis by Menn, Michel et al. (2022).*

METHODOLOGY

Participants: Sixty-seven 10-month-old Dutch infants from the YOUth cohort, each recorded in two EEG sessions 1 week apart (Van der Velde et al., 2019), while listening to the same nursery rhymes as in Menn, Ward et al. (2022).

Pre-processing Pipelines: Developed multiple automated pipelines based on recent infant EEG literature. Varied key parameters, including filter settings, artifact-removal techniques, interpolation methods and epoch thresholds (see *Figure 2*).

Evaluation: We calculated SBC at 4 frequency rates (Stress: 1-3 Hz, Syllabic: 1.7-6 Hz, Phonemic: 5-15 Hz, Total: 1-15 Hz). We computed ICC (3,1) for the 4 SBC measures across pipelines, using both full data and data without outliers (more than ±2 SD from sample mean) to assess the impact of outliers on test-retest reliability.

Raw EEG	BioSemi Prep	Filtering	Bad channel removal	Artifact removal	Interpolation	Re-reference to common average	Clean Epoch check	SBC data
	Re-reference to CZ, Downsample to 512 Hz (Recommended)	Different HP, LP and Notch values	Various criteria	ICA / PCA / ICA + PCA	Weighted avg or Spherical splines	Yes/No	Minimum 30/50/70/90 out of 201	
	✓	HP: 0.1 Hz LP: 40 Hz Notch: 50 Hz	Kurtosis > 10, Variance > 2000, *Abs. amp. > 250,* Flatline > 0.1	RunICA + iMARA	Spherical splines	✓	Minimum 70	

Figure 2*. Overview of the EEG pre-processing pipeline. The top row represents the main pre-processing steps, the middle row are the different parameters tested, and the bottom row shows the settings used for the best performing pipeline.*

RESULTS

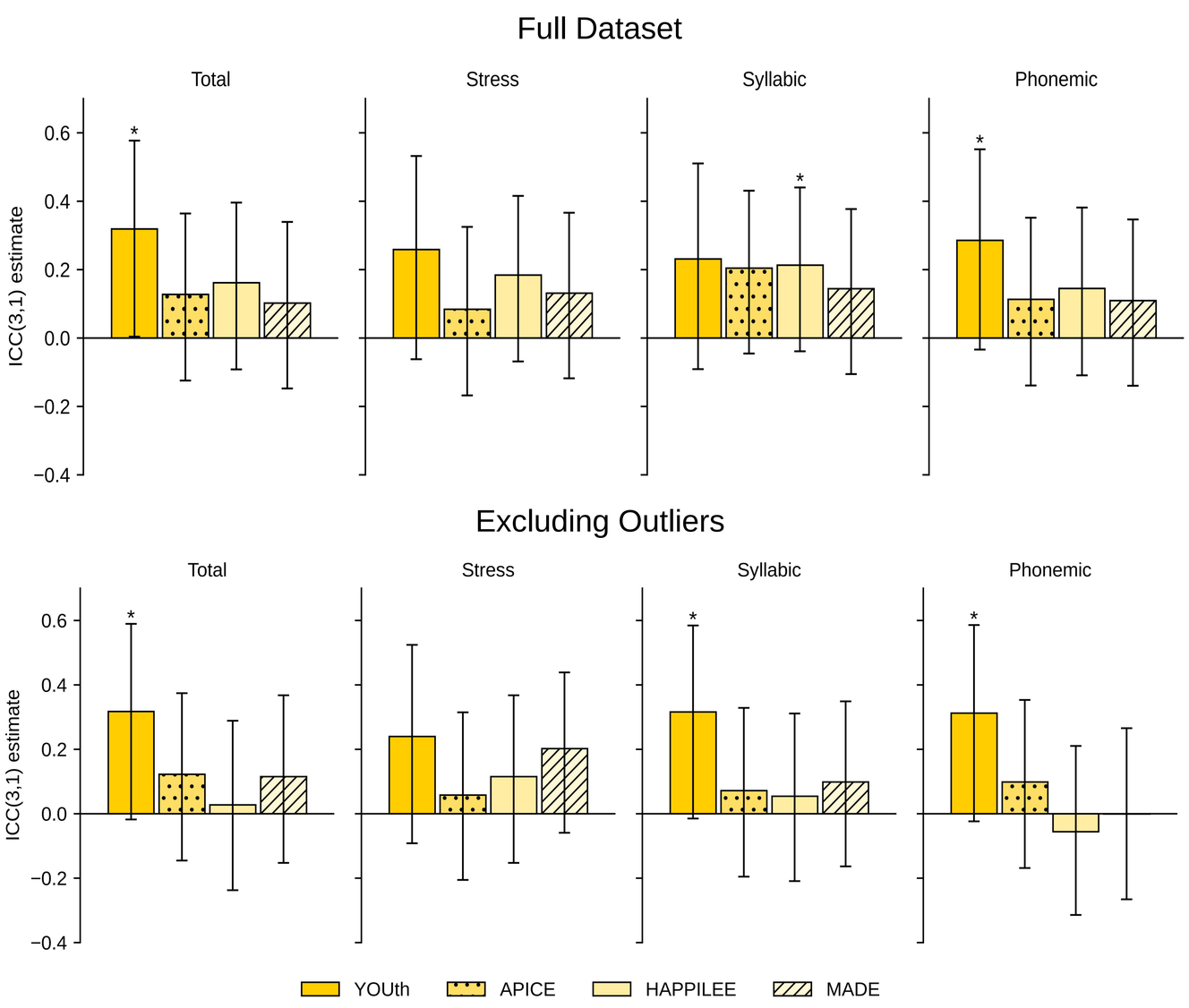


Figure 3: *Bar charts of the test–retest analysis estimates of each pipeline for stress, syllabic, phonemic and total SBC outputs. 95%CIs shown and * marks significant estimates (p < .05). Outliers defined as >±2 SD from the mean.*

SBC Range	Dataset	ICC (95%CI)	F-value	df_1
Total	Full Data	0.32* (0.00, 0.58)	1.93	37
Total	No Outliers	0.32* (-0.02, 0.59)	1.93	33
Stress	Full Data	0.26 (-0.06, 0.53)	1.70	37
Stress	No Outliers	0.24 (-0.09, 0.52)	1.63	35
Syllabic	Full Data	0.23 (-0.09, 0.51)	1.60	37
Syllabic	No Outliers	0.32* (-0.02, 0.58)	1.92	34
Phonemic	Full Data	0.29* (-0.03, 0.55)	1.80	37
Phonemic	No Outliers	0.31* (-0.02, 0.58)	1.91	33

Table 1. *ICC (3,1) results of test-retest reliability analysis for the YOUth pipeline based on the stress, syllabic, phonemic, and total SBC outputs. * p < .05*

DISCUSSION

- POOR but significant ICCs, indicating weak reliability.
- Habituation or learning between sessions?
- Not enough data? (Experiment too short / sample size too small)
- SBC not reliable?
- High variability in infant EEG / low SNR?
- Optimised pipeline still outperformed all the others.

TAKE HOME MESSAGE

Choice of preprocessing pipeline matters, and can influence neural tracking measures.

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